Nephrotic Syndrome Type 12 (NPHS12): A Comprehensive Disease Characterization

Disease: Nephrotic Syndrome Type 12 (NPHS12) **MONDO ID:** MONDO:0014817 · **OMIM:** #616892 · **Causal gene:** *NUP93* **Category:** Mendelian (autosomal recessive) **Report date:** 2026-10-17

Summary

Nephrotic Syndrome Type 12 (NPHS12; MONDO:0014817, OMIM #616892) is an autosomal-recessive, steroid-resistant, kidney-limited podocytopathy caused by biallelic hypomorphic variants in *NUP93*, a gene encoding a central scaffold protein of the inner ring of the nuclear pore complex (NPC). It is one of a growing family of "nucleoporinopathies" — monogenic steroid-resistant nephrotic syndromes (SRNS) caused by mutations in genes encoding NPC subunits (*NUP93, NUP205, NUP107, NUP133, NUP160, NUP85, XPO5*). Clinically, NPHS12 presents in early childhood with proteinuria progressing to the full nephrotic syndrome, resists corticosteroids and other immunosuppressants, shows focal segmental glomerulosclerosis (FSGS) on biopsy, and progresses rapidly to kidney failure.

A critical correction established during this investigation: The MONDO term MONDO:0014817 "nephrotic syndrome, type 12" maps specifically to OMIM:616892 and the *NUP93*-associated disease. The gene *NUP107* — which is superficially similar and causes a closely related nucleoporin SRNS — is responsible for the **distinct** entity NPHS11 (OMIM 616730), not NPHS12. Early iterations of this investigation incorrectly assigned NPHS12 to *NUP107*; ontology verification against EBI OLS4 corrected the gene assignment to *NUP93*. Both genes remain biologically and clinically adjacent (both NPC nucleoporinopathies causing childhood SRNS/FSGS), and much of the shared mechanistic and management narrative applies to both, but the identifiers, onset profile, and molecular partners differ. *NUP93* encodes an **inner-ring** NPC scaffold protein; *NUP107* encodes an **outer-ring (Y-complex)** component.

Mechanistically, mutant NUP93 disrupts nuclear-pore assembly and the NUP93–NUP205 inner-ring interaction. This impairs the NPC-dependent nucleocytoplasmic transport of key podocyte signaling effectors, including SMAD proteins downstream of BMP7 signaling, and (as

shown across the nucleoporin-SRNS family) dysregulates CDC42 and the Hippo-pathway effectors YAP/TAZ. The net result is podocyte injury — foot-process effacement, loss of the glomerular filtration barrier, massive proteinuria, and FSGS. NUP93-SRNS is the most aggressive nucleoporin subtype, with a median age at onset of ~3 years and a median time to kidney failure of ~3.7 years. Because the disease is an intrinsic genetic podocytopathy rather than a circulating-factor-mediated disorder, post-transplant recurrence is rare.

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1. Disease Information

Overview. Nephrotic Syndrome Type 12 is a Mendelian, autosomal-recessive form of steroid-resistant nephrotic syndrome. It belongs to the hereditary podocytopathies and, more specifically, to the subgroup of **nucleoporin-associated SRNS**. Affected children develop heavy proteinuria, hypoalbuminemia, and edema that does not respond to corticosteroid therapy; kidney biopsy typically shows focal segmental glomerulosclerosis (FSGS) with podocyte foot-process effacement. The disease is kidney-limited (non-syndromic) in its classic form.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0014817 ("nephrotic syndrome, type 12")
OMIM (phenotype)	#616892
Causal gene	*NUP93* (OMIM *614351)
HGNC	HGNC:28958
UniProt	Q8N1F7 (NUP93_HUMAN)
Cytoband	16q13
Disease synonym	"NUP93 familial nephrotic syndrome"

Synonyms / alternative names. NPHS12; NUP93-associated steroid-resistant nephrotic syndrome; NUP93 familial nephrotic syndrome; nucleoporin-93 nephrotic syndrome. Within the broader clinical literature it is classified under hereditary/genetic SRNS and FSGS.

Important disambiguation. MONDO:0014817 / OMIM 616892 = **NPHS12** = ***NUP93***. The adjacent entity NPHS11 (OMIM 616730) = ***NUP107***. These should not be conflated despite shared biology (see Finding F008).

Data provenance. Information in this report is derived from **aggregated, disease-level resources** (OMIM, MONDO, UniProt, gnomAD, ClinVar) and from **primary clinical literature** describing individual patient cohorts (whole-exome/whole-genome sequencing case series). No single EHR-derived patient dataset underlies this report; the evidence is literature- and database-aggregated.

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2. Etiology

Primary cause — genetic. NPHS12 is caused by **biallelic (homozygous or compound heterozygous) pathogenic variants in *NUP93***. Braun et al. (2016) established *NUP93* — together with its interaction partner *NUP205* and *XPO5* (exportin-5) — as a hitherto unrecognized monogenic cause of SRNS in eight families:

> "we identify in eight families with SRNS mutations in NUP93, its interaction partner NUP205 or XPO5 (encoding exportin 5) as hitherto unrecognized monogenic causes of SRNS" — PMID: 26878725

Genetic risk factors. The disease requires two deleterious *NUP93* alleles. There are no established common susceptibility loci or modifier-gene associations specific to NPHS12. The principal "risk factor" for an affected child is therefore carrier status of both parents. **Consanguinity** substantially increases risk because it raises the probability of homozygosity for a rare recessive allele; nucleoporin-SRNS is over-represented in consanguineous cohorts (e.g., Saudi Arabian and Egyptian series — PMID: 41141495, PMID: 40402239).

Environmental risk factors. None established. Unlike acquired (idiopathic, immune-mediated) nephrotic syndrome, NPHS12 is a monogenic disorder; environmental exposures, infections, age, and sex are not causal. There is a reported female predominance in small nucleoporin-SRNS case series (see §9), but this is likely a reporting/ascertainment observation rather than a true sex-linked risk.

Protective factors. No genetic or environmental protective factors are established. Theoretically, any residual (hypomorphic) NUP93 function preserved by a milder allele would be "protective" in the sense of attenuating severity, consistent with the recessive, hypomorphic disease model.

Gene–environment interactions. None documented for NPHS12. The disorder is essentially fully genetically determined; disease expression does not appear to require an environmental trigger.

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3. Phenotypes

NPHS12 presents as classic nephrotic syndrome with the following core phenotypes. Frequencies are drawn from nucleoporin-SRNS cohorts; "variable" indicates limited sample sizes.

Phenotype	Type	HPO term	Onset	Severity / frequency
Steroid-resistant nephrotic syndrome	Clinical/ laboratory	HP:0012588	Early childhood (~3 y for NUP93)	Severe; defining feature (100%)
Proteinuria / nephrotic-range proteinuria	Laboratory	HP:0000093	Early childhood	Severe; near-universal
Hypoalbuminemia	Laboratory	HP:0003073	Concurrent with proteinuria	Moderate–severe
Edema	Clinical sign	HP:0000969	Concurrent	Variable
Focal segmental glomerulosclerosis (FSGS)	Histopathology	HP:0000097	At biopsy	~89% of nucleoporin-SRNS biopsies
Podocyte foot-process effacement	Ultrastructural	HP:0031266	At biopsy	Characteristic
Progression to stage 5 CKD / kidney failure	Clinical	HP:0003774	Median ~3.7 y after onset (NUP93)	High

Phenotype characteristics. - **Age of onset:** Early childhood. For *NUP93*-associated SRNS the median age at onset is ~3 years — the earliest and most aggressive among the nucleoporin subtypes:

> "The median ages at onset of NUP85-, NUP93-, NUP107-, NUP133-, NUP160-, and NUP205-associated SRNS are 7, 3, 4.1, 9, 7, and 2 years, respectively" — PMID: 39331077 - **Severity:** Severe and treatment-refractory. - **Progression:** Progressive, typically reaching kidney failure within a few years of onset. One reported *NUP93* patient developed kidney failure just six months after diagnosis at age 1 year 2 months (PMID: 39814977). - **Frequency among affected individuals:** Proteinuria and steroid resistance are universal; FSGS is seen in ~89% of biopsied nucleoporin-SRNS patients.

Quality-of-life impact. Substantial. Progression to dialysis-dependent kidney failure in early childhood imposes major burdens: dialysis or transplantation, growth impairment, chronic medication, dietary restriction, and repeated hospitalization. No NPHS12-specific EQ-5D/SF-36/PROMIS data exist; QoL inference is drawn from the pediatric kidney-failure literature.

Extrarenal phenotypes. Classic NPHS12 is kidney-limited. (Note: the *NUP107*/*NUP133* nucleoporinopathies can extend to the syndromic Galloway–Mowat phenotype with microcephaly and brain anomalies — relevant to the adjacent NPHS11, not to kidney-limited NPHS12.)

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4. Genetic / Molecular Information

Causal gene. ***NUP93*** (nucleoporin 93 kDa).

Attribute	Value
Gene symbol	*NUP93*
HGNC	HGNC:28958
OMIM (gene)	*614351
UniProt	Q8N1F7
Cytoband	16q13
Protein	Nucleoporin NUP93, a scaffold component of the NPC inner ring
RefSeq transcript	NM_014669.5

Population constraint (gnomAD). *NUP93* shows strong loss-of-function (LoF) constraint at the gene level — **LoF Z = 4.32, LOEUF = 0.639** — indicating depletion of LoF variants in the general population, yet **pLI ≈ 0**, consistent with a gene that is intolerant of complete LoF but causes disease through a **recessive, hypomorphic mechanism** (two partial-loss alleles required). This constraint profile fits a dosage-sensitive scaffold protein essential for NPC assembly.

Pathogenic variants. ClinVar lists ~380 *NUP93* variants (NM_014669.5), of which ~53 are classified pathogenic/likely pathogenic. Reported disease variants are predominantly **missense** affecting protein stability and the NUP93–NUP205 interaction, with additional splice and truncating alleles: - Missense examples c.1655A>G and c.1604A>C were shown to disrupt

NUP93 protein stability by immunohistochemistry of kidney biopsy, with ultrastructural "severe vacuolization and deformed nucleus in podocytes, torn and dissolved glomerular basement membrane, and diffuse foot process effacement" — PMID: 39814977.

Variant classification. Per ACMG/AMP criteria, disease alleles are classified pathogenic or likely pathogenic; many novel variants reported in consanguineous cohorts are supported by segregation and functional/in-silico evidence (PMID: 38650033, PMID: 40402239).

Allele frequency. Pathogenic *NUP93* alleles are individually rare (consistent with strong LoF constraint); gnomAD shows depletion of LoF variants. No common high-frequency pathogenic allele is established, though founder alleles may exist in consanguineous populations.

Somatic vs germline. Germline exclusively. No somatic/cancer role.

Functional consequence. Loss of function / hypomorphic, acting through impaired NPC assembly and disrupted NUP93–NUP205 inner-ring scaffolding; dominant-negative mechanisms are not established.

Modifier genes. None validated specific to NPHS12. The broader nucleoporin network (*NUP205*, *XPO5*) is mechanistically linked but represents separate causal loci rather than modifiers.

Epigenetics / chromosomal abnormalities. No disease-specific DNA-methylation or histone signatures reported. NPHS12 is a single-gene disorder; large-scale chromosomal abnormalities are not a typical cause, though rare copy-number events affecting the locus are theoretically possible.

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5. Environmental Information

NPHS12 has **no established environmental, lifestyle, or infectious contributors**. It is a monogenic, fully genetically determined podocytopathy. Toxin exposure, pollution, occupational exposures, smoking, diet, alcohol, and infectious agents are **not applicable** as causes. (Environmental and immune factors dominate *acquired*/idiopathic nephrotic syndrome, which is a distinct differential diagnosis — see §10.)

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic hypomorphic *NUP93* variants** (germline) **lead to** reduced/destabilized NUP93 scaffold protein in podocytes.
2. Destabilized NUP93 **impairs** the NUP93–NUP205 inner-ring interaction and **disrupts** proper nuclear pore complex (NPC) assembly in the nuclear envelope (demonstrated via protein-stability and ultrastructural studies).
3. Defective NPCs **result in** abnormal nucleocytoplasmic transport of signaling effectors in podocytes. *(Branch point.)*
4. **Branch A — SMAD/BMP7 axis (inferred for NUP93 from the Braun 2016 framework):** impaired nuclear import of SMAD proteins **abrogates** BMP7-induced SMAD signaling required for podocyte homeostasis.
5. **Branch B — CDC42 dysregulation (demonstrated across nucleoporin family):** loss of outer-ring/scaffold nucleoporins **dysregulates** CDC42, an effector of SRNS pathogenesis (PMID: 30179222; *NUP160* mouse model PMID: 40298220).
6. **Branch C — Hippo/YAP-TAZ axis (demonstrated for NUP205/NUP107):** impaired NUP-dependent nuclear import **dysregulates** YAP/TAZ transcriptional activity critical for maintaining the glomerular filter (PMID: 37565816).
7. Converging signaling dysregulation **leads to** podocyte injury — cytoskeletal disruption and cellular stress (vacuolization, deformed nuclei, extended/dilated ER observed ultrastructurally).
8. Podocyte injury **results in foot-process effacement** and breakdown of the slit-diaphragm/glomerular filtration barrier.
9. Loss of filtration-barrier integrity **causes** massive **proteinuria → hypoalbuminemia → edema** (clinical nephrotic syndrome), which is **steroid-resistant** because the defect is intrinsic/structural rather than immune-mediated.
10. Ongoing podocyte loss **leads to focal segmental glomerulosclerosis (FSGS)** and progressive nephron dropout.
11. Progressive glomerulosclerosis **results in** chronic kidney disease and **end-stage kidney failure** (median ~3.7 y after onset for NUP93).

Mechanistic detail by category

- **Molecular pathways:** BMP7–SMAD signaling (nuclear import of SMADs); CDC42 (Rho-family GTPase) signaling; Hippo pathway effectors YAP/TAZ. All converge on podocyte cytoskeletal and transcriptional homeostasis.
- **Cellular processes:** Nuclear pore complex assembly (GO:0051292); nucleocytoplasmic transport (GO:0006913); protein import into nucleus (GO:0006606); podocyte injury and apoptosis.
- **Protein dysfunction:** Loss-of-function/destabilization of the NUP93 scaffold; impaired NUP93–NUP205 binding; faulty NPC architecture. Ultrastructurally, mutant NUP93 produces "severe vacuolization and deformed nucleus in podocytes" (PMID: 39814977).
- **Immune involvement:** None primary — distinguishes NPHS12 from immune-mediated nephrotic syndrome and explains immunosuppressant resistance.
- **Tissue damage mechanisms:** Podocyte depletion → glomerular basement membrane disruption → segmental sclerosis/fibrosis.
- **Molecular profiling:** Single-cell transcriptomic/proteomic work in the related *Nup160*-knockout podocyte model shows decreased CDC42 protein with reduced CDC42 activity despite normal/elevated transcript — illustrating a post-transcriptional mechanism (PMID: 40298220).

Ontology suggestions

- **GO biological process:** nuclear pore complex assembly (GO:0051292), nucleocytoplasmic transport (GO:0006913), protein import into nucleus (GO:0006606), nephron development (GO:0072006).
- **GO cellular component:** nuclear pore (GO:0005643), nuclear envelope (GO:0005635), nuclear pore inner ring.
- **CL cell type:** podocyte / glomerular visceral epithelial cell (CL:0000653).
- **UBERON anatomy:** renal glomerulus (UBERON:0000074), kidney (UBERON:0002113).

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7. Anatomical Structures Affected

Organ level. Primary organ: **kidney** (UBERON:0002113), specifically the **renal glomerulus** (UBERON:0000074). Secondary/systemic consequences of nephrotic syndrome include edema, hyperlipidemia, thromboembolic risk, and infection susceptibility; end-stage disease affects the whole organism through uremia. Body system: **urinary/renal system**.

Tissue and cell level. The target cell is the **podocyte** (glomerular visceral epithelial cell, CL:0000653) — a specialized epithelial cell forming the filtration barrier. The glomerular basement membrane and slit diaphragm are secondarily disrupted. Although NUP93 is ubiquitously expressed, the clinical phenotype is podocyte-predominant, reflecting the podocyte's unique vulnerability to NPC/transport dysfunction.

Subcellular level. The **nuclear envelope** (GO:0005635) and **nuclear pore complex** (GO:0005643) are the primary subcellular targets. Secondary ultrastructural abnormalities include deformed nuclei, cytoplasmic vacuolization, and extended/dilated endoplasmic reticulum in podocytes (PMID: 39814977).

Localization / lateralization. Bilateral and diffuse (both kidneys); glomerular involvement is focal and segmental at the histologic level (FSGS pattern) but systemic across the glomerular population.

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8. Temporal Development

Onset. Early-childhood onset; insidious to subacute presentation with proteinuria/edema. For *NUP93* specifically, median age at onset ≈ **3 years** (PMID: 39331077).

Progression. Progressive and typically rapid. NUP93-SRNS is the **most aggressive** nucleoporin subtype, with a **median time to kidney failure of ~3.7 years** from onset. Disease stages run from early proteinuria → nephrotic syndrome → CKD stages → end-stage kidney failure. One case progressed to kidney failure within six months of diagnosis (PMID: 39814977).

Course and duration. Chronic, lifelong; relentlessly progressive without spontaneous remission. Because the disease is intrinsic, immunosuppression-induced remission is unexpected.

Critical periods. The key intervention window is at/near diagnosis — early genetic diagnosis prevents futile immunosuppression and guides transplant planning. There is no known "protective" developmental window.

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9. Inheritance and Population

Inheritance. Autosomal recessive (biallelic, homozygous or compound heterozygous) — PMID: 26878725.

Penetrance / expressivity. Penetrance appears high in biallelic carriers; expressivity is variable with respect to age at kidney failure, likely reflecting allele severity (hypomorphic spectrum).

Consanguinity and founder effects. Consanguinity is a major contributing factor, enriching nucleoporin-SRNS in Middle Eastern, North African, and South/East Asian cohorts (Saudi PMID: 41141495; Egyptian PMID: 40402239; Chinese PMID: 38650033). Population-specific founder alleles plausibly exist but are not catalogued as a single dominant allele.

Carrier frequency. Not precisely established; expected to be low given strong LoF constraint and rarity of pathogenic alleles in gnomAD.

Epidemiology. NPHS12 is **ultra-rare**. Nucleoporin variants are described as "extremely rare" causes of hereditary SRNS (PMID: 38650033). No reliable prevalence/incidence figures per 100,000 exist; cases are reported as individual families or small series. Within monogenic SRNS cohorts, nucleoporin genes collectively account for a small minority of diagnoses, and *NUP93* is one member of that minority.

Sex ratio. Small case series note a **female predominance** (e.g., all four nucleoporin-SRNS patients in PMID: 38650033 were female), but numbers are too small to establish a true skewed sex ratio; recessive inheritance predicts equal sex distribution.

Age distribution. Affected individuals are predominantly young children (onset ~3 y; kidney failure often in childhood).

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10. Diagnostics

Clinical/laboratory tests. - **Urinalysis:** nephrotic-range proteinuria (HP:0000093); urine protein/creatinine ratio. - **Serum:** hypoalbuminemia (HP:0003073), hyperlipidemia; renal function (creatinine, eGFR) to stage CKD. - **Biopsy/histopathology:** FSGS on light microscopy; **electron microscopy** shows diffuse podocyte **foot-process effacement**, and in NUP93 disease, podocyte vacuolization, deformed nuclei, GBM disruption, and dilated ER (PMID: 39814977). Immunohistochemistry can show reduced/destabilized NUP93 protein.

Genetic testing (the definitive diagnostic). - **Recommended approach:** Next-generation sequencing using a **targeted SRNS/podocytopathy gene panel** that includes *NUP93* and other nucleoporin genes — "Next-generation sequencing using a targeted gene panel is recommended" (PMID: 39331077). - **WES/WGS:** High-yield in early-onset SRNS. Whole-genome sequencing identified a monogenic cause in 56.1% of early-onset SRNS families, including *NUP93* among causal genes (PMID: 40402239); exome sequencing yields 40–45% in consanguineous/pediatric kidney-failure cohorts (PMID: 41141495, PMID: 37850020). - **Confirmatory:** Sanger sequencing with family segregation; minigene/cDNA assays for splice variants.

Clinical criteria. Diagnosis of SRNS requires nephrotic syndrome with failure to achieve remission after an adequate corticosteroid course; the genetic diagnosis is established by identifying biallelic pathogenic *NUP93* variants.

Differential diagnosis. Other genetic SRNS/FSGS causes must be distinguished — *NPHS1*, *NPHS2*, *WT1*, *PLCE1*, *LAMB2*, *PAX2* (PMID: 31001663), *COL4A3-5* (PMID: 36847718), *SMARCAL1*, and other nucleoporins (*NUP107/NPHS11*, *NUP205*, *NUP160*, *NUP85*, *NUP133*). Immune-mediated/idiopathic steroid-sensitive nephrotic syndrome and secondary FSGS are also differentials; genetic testing resolves these and avoids unnecessary immunosuppression.

Screening. Cascade carrier testing of relatives and genetic counseling are appropriate once a familial variant is known; prenatal/preimplantation testing is feasible for known biallelic variants.

11. Outcome / Prognosis

Renal prognosis — poor. NPHS12 progresses to kidney failure, typically in childhood. NUP93-SRNS has the shortest median time to kidney failure among nucleoporin subtypes (~3.7 years), and is described as rapidly progressive to ESKD (PMID: 39331077, PMID: 38650033).

Mortality/survival. No disease-specific survival statistics; prognosis is dominated by kidney failure and is favorable for survival with renal replacement therapy (dialysis/transplant). The disease is kidney-limited, so extrarenal mortality drivers are absent (unlike syndromic nucleoporinopathies).

Morbidity/QoL. High morbidity from childhood kidney failure: dialysis dependence, transplant, growth and developmental impact, and chronic immunosuppression burden post-transplant.

Complications. Nephrotic-syndrome complications (infection, thromboembolism, hyperlipidemia, malnutrition) plus CKD/ESKD complications.

Transplant outcome — favorable with low recurrence. Because NPHS12 is an intrinsic genetic podocytopathy (not circulating-factor-mediated), **post-transplant recurrence is rare**. A critical re-analysis found only one plausible biallelic *NUP93* recurrence case; overall "post-transplantation recurrence remains very rare in patients with genetic SRNS" (PMID: 34031708). Reported recurrence for nucleoporin-SRNS is ~12.5%, essentially limited to NUP93.

Prognostic factors. Earlier onset and more deleterious (truncating/strongly destabilizing) alleles predict faster progression. Genotype (NUP93 among the most aggressive nucleoporin subtypes) is itself prognostic.

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12. Treatment

No curative or disease-specific therapy exists. Management is supportive and focused on renal replacement.

Pharmacotherapy. - **Corticosteroids and immunosuppressants are ineffective** — the disease is steroid-resistant by definition, and "Most affected individuals are resistant to immunosuppressants" (PMID: 39331077). Establishing a genetic diagnosis is important precisely to **avoid** futile, toxic immunosuppression. - **RAAS blockade** (ACE inhibitors / ARBs) is used as **antiproteinuric/nephroprotective** supportive therapy, as in other genetic

podocytopathies; partial/complete remission on RAAS antagonists has been reported in a minority of monogenic proteinuria patients generally (PMID: 36847718) (NCIT: angiotensin-converting enzyme inhibitor therapy). - **Supportive care:** diuretics for edema, lipid management, anticoagulation prophylaxis where indicated, nutrition, infection prevention.

Renal replacement therapy. - **Dialysis** at end-stage kidney failure (NCIT: dialysis). - **Kidney transplantation** — treatment of choice for ESKD in genetic SRNS, with generally good graft outcomes and low recurrence risk (PMID: 34031708) (NCIT: kidney transplantation).

Advanced/experimental therapeutics. No approved gene, cell, or RNA therapies for NPHS12. Mechanistic insights (CDC42, YAP/TAZ, SMAD/BMP7 axes) suggest future targeted strategies but none are clinically validated.

Pharmacogenomics. Not applicable/established.

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13. Prevention

Primary prevention of this monogenic disorder is limited to **reproductive genetic counseling**: carrier screening of at-risk families (especially consanguineous couples), prenatal diagnosis, and preimplantation genetic testing once a familial biallelic *NUP93* variant is identified. There is no vaccine, lifestyle, or environmental prevention because the disease is not environmentally triggered.

Secondary prevention — early genetic diagnosis via NGS panels enables prompt, appropriate management: avoiding ineffective immunosuppression, initiating nephroprotection (RAAS blockade), and planning transplantation. Genetic diagnosis "is crucial to inform clinical decision making, genetic counseling, transplantation strategy and prenatal diagnosis" (PMID: 40402239).

Tertiary prevention — CKD complication management, nutrition, infection and thrombosis prophylaxis in the nephrotic state, and optimization of transplant outcomes.

Counseling. Autosomal-recessive recurrence risk of 25% per pregnancy for carrier couples; cascade testing of relatives is recommended.

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14. Other Species / Natural Disease

Taxonomy / orthologs. *NUP93* is evolutionarily conserved with clear orthologs in mouse (*Nup93*), rat, zebrafish, and other vertebrates (NPC subunits are deeply conserved from yeast to human). No naturally occurring companion-animal or wildlife disease specifically attributed to *NUP93* nephrotic syndrome is catalogued in OMIA at this time.

Comparative biology. The podocyte filtration barrier and NPC biology are conserved across vertebrates, making model-organism findings translatable. Within the nucleoporin-SRNS family, zebrafish and Xenopus morphant/knockdown studies of related nucleoporins (*nup107*, *nup133*) reproduce glomerular/podocyte defects (PMID: 30427554, PMID: 30179222), demonstrating conserved disease mechanisms.

Zoonotic potential. None — not an infectious disease.

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15. Model Organisms

Direct *Nup93* models. A dedicated *Nup93* podocyte-specific mouse recapitulating NPHS12 was not identified in this investigation; however, the nucleoporin-SRNS family is well-modeled and the mechanisms are shared.

Family-level models (mechanistically informative for NPHS12). - **Mouse:** A **podocyte-specific *Nup160* knockout mouse** (CRISPR/Cas9 + Cre/loxP) develops progressive proteinuria and podocyte foot-process effacement — faithfully recapitulating the human nephrotic phenotype — and demonstrates CDC42 protein/activity dysregulation via single-cell transcriptomics and proteomics (PMID: 40298220). This is the strongest in-vivo demonstration of the nucleoporin → CDC42 → podocyte-injury mechanism. - **In-vitro podocytes:** CRISPR/Cas9 knockout of *NUP107/NUP85/NUP133* in immortalized human podocytes activates CDC42 (PMID: 30179222); interactome studies in podocytes link *NUP107/133/205/XPO5* to YAP/TAZ regulation (PMID: 37565816). - **Zebrafish/Xenopus:** Morpholino knockdown of *nup107* and *nup133* produces hypoplastic glomeruli and podocyte foot-process fusion, rescuable by wild-type but not mutant human mRNA — establishing causality (PMID: 30427554).

Phenotype recapitulation. These models reproduce the core human phenotype (proteinuria, foot-process effacement, glomerular injury) and the molecular mechanism (CDC42/YAP-TAZ dysregulation downstream of NPC disruption). **Limitation:** they target *paralogous* nucleoporins rather than *NUP93* itself, and the kidney-limited human phenotype may not be fully captured where models cause broader developmental defects.

Resources. MGI (mouse *Nup93*), ZFIN (zebrafish *nup* genes), Alliance of Genome Resources for orthology; immortalized human podocyte lines and iPSC-derived podocytes/kidney organoids are suitable in-vitro platforms for future *NUP93*-specific modeling.

Mechanistic Model (synthesis)

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Biallelic hypomorphic NUP93 variants (germline, AR)
            │  (destabilized NUP93 scaffold protein)
            ▼
Impaired NUP93–NUP205 inner-ring binding → defective NPC assembly
            │  (nuclear envelope / nuclear pore defect; GO:0051292)
            ▼
Disrupted nucleocytoplasmic transport in podocytes (GO:0006913)
      ├── Branch A: ↓ SMAD nuclear import → impaired BMP7–SMAD signaling (inferred)
      ├── Branch B: CDC42 dysregulation (demonstrated in NUP160/107/85/133 models)
      └── Branch C: YAP/TAZ mislocalization (demonstrated for NUP205/107)
            │
            ▼
Podocyte injury (vacuolization, deformed nuclei, dilated ER, cytoskeletal failure)
            │
            ▼
Foot-process effacement → loss of glomerular filtration barrier (CL:0000653, UBERON:0000074)
            │
            ▼
Nephrotic-range proteinuria → hypoalbuminemia → edema  [STEROID-RESISTANT]
            │
            ▼
Focal segmental glomerulosclerosis (FSGS, ~89%)
            │
            ▼
Rapid progression → end-stage kidney failure (median ~3.7 y; NUP93 = most aggressive)
            │
            ▼
Kidney transplant → LOW recurrence (intrinsic genetic podocytopathy)
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Evidence Base

PMID	Study	How it supports the findings
26878725	Braun et al. 2016 — *Mutations in nuclear pore genes NUP93, NUP205 and XPO5 cause SRNS*	**Landmark** — establishes *NUP93* (with *NUP205*, *XPO5*) as monogenic SRNS cause; defines NPHS12's causal gene and NPC/SMAD mechanism
39331077	Yao 2025 — *Nucleoporin-associated SRNS* (review)	Median onset ages (NUP93 ≈ 3 y), aggressive course, FSGS ~89%, immunosuppressant resistance, panel-based diagnosis, low transplant recurrence
39814977	2025 cohort — NUP85/93/107/160 SRNS	Direct NUP93 patient: kidney failure 6 mo after diagnosis; missense variants destabilize NUP93; podocyte vacuolization, deformed nuclei, GBM disruption, foot-process effacement
38650033	2024 Chinese cohort — NUP93/107/160 SRNS	Biallelic variants cause severe early-onset SRNS → rapid ESKD; FSGS; expands variant spectrum
30179222	Braun et al. 2018 — *Multiple NPC components cause nephrotic syndrome*	CRISPR KO of nucleoporins in podocytes activates CDC42 — shared mechanism
40298220	2025 — Podocyte-specific *Nup160* KO mouse	In-vivo model: proteinuria + foot-process effacement; CDC42 dysregulation (single-cell omics)
37565816	2023 — NUP205 regulates YAP/TAZ	Links NPC dysfunction to Hippo-effector mislocalization in podocytes
34031708	2021 — Re-analysis of post-transplant recurrence in genetic NS	One plausible biallelic *NUP93* recurrence; genetic SRNS recurrence very rare
40402239	2025 — WGS Egyptian SRNS cohort	Monogenic cause in 56.1% of families incl. *NUP93*; value of genetic diagnosis
41141495	2025 — Saudi pediatric kidney disease exome cohort	Consanguinity (54%), founder effects, genetic diagnostic yield context
26411495	Miyake et al. 2015 — *NUP107* SRNS	Defines the **adjacent** entity (NPHS11/*NUP107*) — key for disambiguation from NPHS12
30427554	2018 — *NUP133* GAMOS, zebrafish model	Conserved NPC → podocyte mechanism; rescue by WT not mutant mRNA

Limitations and Knowledge Gaps

1. **Gene-assignment correction.** This investigation initially (iterations 1–4) misassigned NPHS12 to *NUP107*; ontology verification (EBI OLS4) corrected it to *NUP93* (MONDO:0014817 → OMIM 616892). Readers and downstream knowledge bases must use the corrected *NUP93* assignment. Several cited mechanistic papers describe *paralogous* nucleoporins (*NUP107/160/205*) rather than *NUP93* directly; mechanism is inferred by family analogy where NUP93-specific data are lacking.
2. **Rarity.** NPHS12 is ultra-rare; most evidence is from small case series and consanguineous cohorts. No population-level prevalence/incidence, penetrance quantification, or carrier-frequency estimates exist.
3. **Branch-specific uncertainty.** The SMAD/BMP7 branch is inferred for NUP93 from the original Braun framework; CDC42 and YAP/TAZ branches are best demonstrated in *other* nucleoporins. Which branch dominates in NUP93 podocytes is not firmly established.
4. **No dedicated *NUP93* animal model** was identified; recapitulation evidence relies on paralog models (*Nup160* mouse; *nup107/133* fish/frog).
5. **No disease-specific QoL, mortality, or natural-history registry data**; prognosis is inferred from pooled nucleoporin-SRNS and genetic-SRNS literature.
6. **No veterinary/natural-disease counterpart** catalogued.

Proposed Follow-up Experiments / Actions

1. **Confirm and propagate the corrected identifier mapping** (NPHS12 = MONDO:0014817 = OMIM 616892 = *NUP93*; NPHS11 = OMIM 616730 = *NUP107*) throughout the knowledge base, with an explicit disambiguation note.
2. **Generate a podocyte-specific *Nup93* conditional-knockout (or hypomorphic knock-in) mouse** to test whether NUP93 disease proceeds through CDC42, YAP/TAZ, and/or SMAD branches, mirroring the *Nup160* mouse design.
3. **iPSC-derived podocyte / kidney-organoid models** carrying patient *NUP93* variants (e.g., c.1655A>G, c.1604A>C) to assay NPC assembly, nucleocytoplasmic transport of SMAD/YAP/TAZ, and CDC42 activity.
4. **Genotype–phenotype meta-analysis** aggregating published *NUP93* patients to quantify onset, time-to-kidney-failure, and allele-severity correlations with adequate power.

5. **Evaluate CDC42/Rho-pathway or YAP-TAZ-directed small molecules** in NUP93 podocyte models as candidate targeted therapies.
6. **Prospective transplant-outcome registry** for genetic nucleoporin-SRNS to firm up the ~12.5% NUP93 recurrence signal.
7. **Deep-variant curation** of *NUP93* in ClinVar/gnomAD and consanguineous cohorts to identify founder alleles and refine ACMG classifications.

Report compiled from aggregated database resources (MONDO, OMIM, UniProt, gnomAD, ClinVar, HGNC) and primary literature. Evidence source types span human clinical cohorts, model-organism studies, and in-vitro functional work, as indicated per claim.
